Technical University of Denmark

Advanced machine learning (02460)

Mock exam, spring 2024

Exam duration: 2 hours.

Aid: Only written works of reference are permitted.

IMPORTANT: Only information written on the pages 2–12 will be evaluated.

Info about you: *You must write your student number on all pages (bottom-left).*

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Part I
Generative models

Exercise A Probabilistic PCA

Consider a probabilistic PCA model with latent variable $z \in \mathbb{R}$ and observed variable $\mathbf{x} \in \mathbb{R}^2$. Let z follow a standard Gaussian distribution, i.e., $p(z) = \mathcal{N}(z|0,1)$, and the distribution of $\mathbf{x}$ be given by

$$p(\mathbf{x}|z) = \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix}1\\2\end{bmatrix} z + \begin{bmatrix}1\\-1\end{bmatrix}, 2 \cdot \mathbf{I}_2\right), \tag{1}$$

where $\mathbf{I}_2 \in \mathbb{R}^{2\times 2}$ is the identity matrix.

Question A.1: What is the marginal distribution of x?

A ☐ $p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}\right)$

B ☐ $p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}\right)$

C ☐ $p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}3 & 2\\2 & 6\end{bmatrix}\right)$

D ☐ $p(\mathbf{x}) = \mathcal{N}\left(\mathbf{x} \middle| \begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}3 & 2\\2 & 6\end{bmatrix}\right)$

Give a brief argument for your answer.

A

Exercise B Normalizing Flows

Consider a normalising flow with random variables $\mathbf{u}, \mathbf{x} \in \mathbb{R}^2$, where the base distribution is given by

$$p_{\mathrm{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u}|\mathbf{0}, \mathbf{1}) \tag{2}$$

and the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is given by

$$T(\mathbf{u}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{u} + \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \tag{3}$$

The normalising flow model is driven by the model $\mathbf{u} \sim p_{\mathrm{u}}(\mathbf{u})$ and $\mathbf{x} = T(\mathbf{u})$, and the density of $\mathbf{x}$ is denoted $p_{\mathrm{x}}(\mathbf{x})$.

Recall that the probability density function for the k-variate normal distribution is given by

$$f(\mathbf{y}|\mu, \mathbf{\Sigma}) = \frac{1}{\sqrt{(2\pi)^k |\mathbf{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mu)^T \mathbf{\Sigma}^{-1} (\mathbf{y} - \mu)\right), \tag{4}$$

where $\mathbf{y} \in \mathbb{R}^k$ is the random vector, $\mu \in \mathbb{R}^k$ mean vector and $\mathbf{\Sigma} \in \mathbb{R}^{k \times k}$ is the covariance matrix (symmetric and positive definite).

Question B.1: What is $p_{\mathrm{x}}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$?

A ☐ $p_{\mathrm{x}}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \frac{1}{2\pi}$

B ☐ $p_{\mathrm{x}}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \frac{1}{4\pi}$

C ☐ $p_{\mathrm{x}}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \frac{1}{8\pi}$

D ☐ $p_{\mathrm{x}}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \frac{1}{16\pi}$

Give a brief argument for your answer.

Exercise C Denoising Diffusion Probabilistic Models

Consider a DDPM model of two-dimensional real-valued data with $T = 5$ latent variables, i.e., where $\mathbf{x}_0 \in \mathbb{R}^2$ is that data variable and $\mathbf{x}_1, \ldots, \mathbf{x}_5 \in \mathbb{R}^2$ are the latent variables. Let the variance schedule be given by $\beta_1 = 10^{-4}$, $\beta_2 = 10^{-3}$, $\beta_3 = 10^{-2}$, $\beta_4 = 10^{-1}$ and $\beta_5 = \frac{3}{4}$.

Recall that the forward process in the DDPM is a Markov chain that gradually adds Gaussian noise to the data using, i.e.,

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}) \tag{5}$$

Question C.1: Assuming that $\mathbf{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\mathbf{x}_4 = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$, what is the expression for $q(\mathbf{x}_5|\mathbf{x}_4, \mathbf{x}_0)$?

$q(\mathbf{x}_5|\mathbf{x}_4, \mathbf{x}_0) =$ _____

Give a brief argument for your answer.

Part II

Representation learning and identifiability

Exercise D Pull-back metrics

Consider the generative model

$$f(\mathbf{x}) = \begin{pmatrix} 3x_1^4 + x_2^2 \\ 4x_1^3 + x_2 \\ x_1 + x_2 \end{pmatrix} \tag{6}$$

as well as the following four matrices

$$\mathbf{G}_{\mathbf{x}}^{(a)} = \begin{pmatrix} 9x_1^8 + 16x_1^6 + x_1^2 & 3x_1^4x_2^2 + 4x_1^3x_2 + x_1x_2 \\ 3x_1^4x_2^2 + 4x_1^3x_2 + x_1x_2 & x_2^4 + 2x_2^2 \end{pmatrix}$$

$$\mathbf{G}_{\mathbf{x}}^{(b)} = \begin{pmatrix} 144x_1^6 + 144x_1^4 + 1 & 24x_1^3x_2 + 12x_1^2 + 1 \\ 24x_1^3x_2 + 12x_1^2 + 1 & 4x_2^2 + 2 \end{pmatrix}$$

$$\mathbf{G}_{\mathbf{x}}^{(c)} = \begin{pmatrix} 144x_1^6 + 4x_2^2 & 144x_1^5 + 2x_2 & 12x_1^3 + 2x_2 \\ 144x_1^5 + 2x_2 & 144x_1^4 + 1 & 12x_1^2 + 1 \\ 12x_1^3 + 2x_2 & 12x_1^2 + 1 & 1 \end{pmatrix}$$

$$\mathbf{G}_{\mathbf{x}}^{(d)} = \begin{pmatrix} 9x_1^8 + x_2^4 & 12x_1^7 + x_2^3 & 3x_1^5 + x_2^3 \\ 12x_1^7 + x_2^3 & 16x_1^6 + x_2^2 & 4x_1^4 + x_2^2 \\ 3x_1^5 + x_2^3 & 4x_1^4 + x_2^2 & x_1^2 + x_2^2 \end{pmatrix}$$

Question D.1: The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{J}_{\mathbf{x}}^{\intercal}\mathbf{J}_{\mathbf{x}}$, where $\mathbf{J}_{\mathbf{x}}$ is the Jacobian of f. Which one of the following statements is true?

A ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(a)}$.

B ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(b)}$.

C ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(c)}$.

D ☐ The pull-back metric is $\mathbf{G}_{\mathbf{x}} = \mathbf{G}_{\mathbf{x}}^{(d)}$.

Give a brief argument for your answer.

D

Exercise E Curve energy

Consider the generative model $\mathbf{y} = f(\mathbf{x})$, where

$$f(\mathbf{x}) = \begin{pmatrix} x_1 \\ x_2 \\ x_1^2 + x_2^2 \end{pmatrix}. \tag{7}$$

Furthermore, consider the two latent curves

$$c_1(t) = t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad c_2(t) = t \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{8}$$

Question E.1: Derive the pull-back metric $\mathbf{G}_\mathbf{x} = \mathbf{J}_\mathbf{x}^\intercal \mathbf{J}_\mathbf{x}$, where $\mathbf{J}_\mathbf{x}$ denotes the Jacobian of f at $\mathbf{x}$.

Question E.2: Let the energy of a curve be given by

$$\mathcal{E}(c) = \int_0^1 \dot{c}(t)^\intercal \mathbf{G}_{c(t)} \dot{c}(t) \mathrm{d}t, \tag{9}$$

where $\dot{c}(t) = \frac{\mathrm{d}c(t)}{\mathrm{d}t}$ is the curve derivative. Evaluate the energy of the curves c_1 and c_2.

$\mathcal{E}(c_1) =$ _____ $\mathcal{E}(c_2) =$ _____

Exercise F Integration

Consider the metric over $\mathbb{R}^2$

$$\mathbf{G}_{\mathbf{x}} = \begin{pmatrix} x_1^2 & 0 \\ 0 & 2x_2^2 \end{pmatrix}. \tag{10}$$

Question F.1: Derive the volume measure $\mathrm{d}\mathcal{M}(\mathbf{x}) = \sqrt{\det \mathbf{G}_{\mathbf{x}}}\mathrm{d}\mathbf{x}$.

$\mathrm{d}\mathcal{M}(\mathbf{x}) =$ _____

Question F.2: Evaluate the integral $I = \int_{\Omega} x_1 + x_2 \mathrm{d}\mathcal{M}(\mathbf{x})$, where $\Omega = [-1, 1] \times [-1, 1]$.

$I =$ _____

F

Exercise G Information geometry

In information geometry, the *Bregman divergence* for comparing two distributions p and q is given as

$$D_F(p, q) = F(p) - F(q) - \langle \nabla F(q), p - q \rangle, \tag{11}$$

for some function $F : \mathcal{H} \to \mathbb{R}$, where $\mathcal{H}$ denotes the space of distributions.

Consider the two probability vectors

$$p = \begin{pmatrix} p_1 \\ \vdots \\ p_D \end{pmatrix} \quad \text{and} \quad q = \begin{pmatrix} q_1 \\ \vdots \\ q_D \end{pmatrix}, \tag{12}$$

where $p_i > 0$, $q_i > 0$, and $\sum_i p_i = \sum_i q_i = 1$.

Let F evaluate the squared Euclidean norm

$$F(p) = \|p\|^2 = \sum_{i=1}^{D} p_i^2. \tag{13}$$

Question G.1: Show that the Bregman divergence D_F is then the squared Euclidean distance

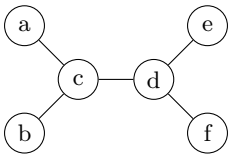
$$\|p - q\|^2 = \sum_{i=1}^{D} (p_i - q_i)^2. \tag{14}$$

G

Part III
Graph models

Exercise H Weisfieler-Lehman test

Consider the following graph



Question H.1: We run the Weisfeiler-Lehman algorithm for a sufficiently large number of rounds on this graph, and examine the final hash-values assigned to each node:

- Assign initial labels
$$l_v^{(0)} = d_v$$
- Iteratively assign new labels by hashing the multi-set within neighborhood
$$l_v^{(i)} = \text{HASH}(\{\{l_u^{(i-1)} \forall u \in \mathcal{N}(v)\}\})$$

Which of the following statements are true. **Select all that apply.**

A ☐ Node a and b will have the same hash value.

B ☐ Node a and c will have the same hash value.

C ☐ Node a and e will have the same hash value.

D ☐ Node e and f will have the same hash value.

E ☐ Node c and d will have the same hash value.

Give a brief argument for your answer.

Exercise I Shallow embeddings

In this exercise we will use the decoder $P(A_{u,v} = 1) = \mathrm{DEC}(\boldsymbol{z}_u, \boldsymbol{z}_v) = \sigma(\boldsymbol{z}_u^\top \boldsymbol{z}_v + b)$, where $\sigma(x) = \frac{1}{1+e^{-x}}$ denotes the sigmoid function. We will use a 2-dimensional embedding space, and refer to the coordinates of the embedding as $\boldsymbol{z}_v = [x, y]^\top$.

Question I.1: Let us assume that node u is embedded as $\boldsymbol{z}_u = [1, 0]^\top$ and that we have $b = 1$. Where could node v be embedded, if the probability of a link between u and v should be $P(A_{u,v} = 1) = 0.5$?

A ☐ On a circle of radius 0.5 around $\boldsymbol{z}_u$, i.e. $(x-1)^2 + y^2 = 0.5^2$

B ☐ On a circle of radius 1 around the origin, i.e. $x^2 + y^2 = 1$

C ☐ On the line $x = -1$

D ☐ On the line $y = 0$

E ☐ None of the above

Show the main steps of your derivation.

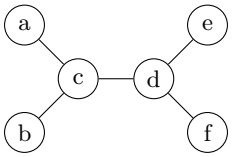
Question I.2: Three nodes (u, v, and w) are embedded as $\boldsymbol{z}_u = [2, 0]^\top$, $\boldsymbol{z}_v = [-2, 0]^\top$, and $\boldsymbol{z}_u = [0, 2]^\top$. Determine a value for b such that we have $P(A_{u,w} = 1) = P(A_{v,w} = 1) = 1 - P(A_{u,v} = 1)$.

$b =$ _____

Show the main steps of your derivation.

Exercise J Centrality and graph convolution

Consider the following graph



The following code prints its adjacency matrix and computes its eigendecomposition (the the nodes are ordered *a*–*f*.)

```
>>> print(A)
tensor([[0., 0., 1., 0., 0., 0.],
        [0., 0., 1., 0., 0., 0.],
        [1., 1., 0., 1., 0., 0.],
        [0., 0., 1., 0., 1., 1.],
        [0., 0., 0., 1., 0., 0.],
        [0., 0., 0., 1., 0., 0.]])

>>> l, U = np.linalg.eigh(A)

>>> print(np.round(U, 3))
[[-0.289  0.408 -0.    -0.707 -0.408 -0.289]
 [-0.289  0.408 -0.     0.707 -0.408 -0.289]
 [ 0.577 -0.408 -0.     0.    -0.408 -0.577]
 [-0.577 -0.408  0.     0.     0.408 -0.577]
 [ 0.289  0.408  0.707  0.     0.408 -0.289]
 [ 0.289  0.408 -0.707  0.     0.408 -0.289]]
```

The eigenvector centrality of a node u is defined as

$$e_u = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} e_v$$

Question J.1: What is the eigenvector centrality of node c?

$e_c =$ _____

Circle the entry in the matrix of eigenvectors that you based your answer on.

Question J.2: What is the principal eigenvalue?

$\lambda =$ _____

A graph convolution with a finite-length filter can be written in the vertex domain, and in the spectral domain as:

$$\boldsymbol{y} = \sum_{k=0}^{N} h[k]\boldsymbol{A}^k\boldsymbol{x} = \boldsymbol{U}\left(\sum_{k=0}^{N} h[k]\boldsymbol{\Lambda}^k\right)\boldsymbol{U}^*\boldsymbol{x} \tag{15}$$

Consider a signal $\boldsymbol{x}$ defined on the nodes a–f in the graph above:

$$\boldsymbol{x} = [1, 1, 0, 0, 1, 1]^\top$$

Question J.3: Given the filter

$$h[k] = \begin{cases} 1 & k = 1 \\ 0 & \text{otherwise,} \end{cases}$$

what is the filtered signal $\boldsymbol{y}$?

$\boldsymbol{y} =$ _____

J